

Deep Learning Math 1 · Probability Foundations & Entropy — Notes

Companion to [prob01-foundations.html](#) . Full definitions, expanded proofs, edge cases, and the lemmas the deck uses without proof. Throughout, $\log$ denotes $\log_2$ and entropies are in bits; $\ln$ is the natural logarithm.

1 Why Probability?

The deck's opening claim is that every object of deep learning is a probabilistic object. Slightly more carefully:

- **Data are samples.** A dataset $x_1, \dots, x_n$ is modeled as n independent draws from an unknown distribution P on a sample space Ω (the space of all possible images, sentences, etc.). "Learning" means inferring structure of P — its high-mass regions, its conditionals given a label — from these finitely many draws. The i.i.d. assumption is itself a modeling choice: it fails for time series, curated datasets, or feedback loops, and later lectures (random processes, concentration) treat what happens when it holds or is relaxed.
- **Models are distributions.** A classifier does not output a label; it outputs a probability mass function over labels (a soft label) — the softmax layer exists precisely to turn arbitrary real scores into a valid pmf. A generative model is a distribution over Ω from which we can sample. An RL policy is a conditional distribution over actions given a state.

The lecture's technical goal is one number: the entropy $H(X)$, which summarizes how uncertain a distribution is, together with its exact range $0 \leq H(X) \leq \log M$ and the proof technique (Jensen's inequality) that recurs throughout the course — KL non-negativity in Lecture 2, the ELBO in generative modeling.

2 Probability Review

2.1 Sample space, events, and the event collection

The **sample space** Ω is the set of all possible outcomes of the experiment. An **event** E is a subset of Ω ; the event " E occurred" means the realized outcome ω satisfies $\omega \in E$. The **event collection** $\mathcal{F}$ is a family of subsets of Ω — the events to which we are willing to assign probabilities.

The slides say "a (measurable) subset" and move on; here is what that parenthesis hides. In a fully rigorous treatment $\mathcal{F}$ must be a **σ -algebra**: (i) $\Omega \in \mathcal{F}$; (ii) if $E \in \mathcal{F}$ then the complement $E^c \in \mathcal{F}$; (iii) if $E_1, E_2, \dots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} E_i \in \mathcal{F}$. These closure properties guarantee that logical combinations of legitimate questions ("not E ", " E_1 or E_2 ") are again legitimate questions. When Ω is finite or countable — every example in this lecture — one may always take $\mathcal{F} = 2^{\Omega}$, the collection of all subsets, and measurability is a non-issue. The distinction only bites for uncountable Ω (e.g. $\Omega = [0, 1]$), where not every subset can consistently be assigned a probability; this course stays discrete until differential entropy (Lecture 3).

2.2 Probability measure: the axioms and their first consequences

A **probability measure** is a map $P : \mathcal{F} \rightarrow [0, 1]$ satisfying the Kolmogorov axioms:

Axioms. (i) $P(E) \geq 0$ for every $E \in \mathcal{F}$; (ii) $P(\Omega) = 1$; (iii) countable additivity: if $E_1, E_2, \dots$ are pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$

The two facts quoted on the slides are consequences, not axioms:

- **$P(\emptyset) = 0$.** Take $E_1 = \Omega$ and $E_i = \emptyset$ for $i \geq 2$ in (iii): $1 = P(\Omega) = 1 + \sum_{i \geq 2} P(\emptyset)$, forcing $P(\emptyset) = 0$.
- **Monotonicity ("larger event, larger probability").** If $A \subseteq B$, write $B = A \cup (B \setminus A)$, a disjoint union. Additivity gives $P(B) = P(A) + P(B \setminus A) \geq P(A)$, since the second term is non-negative by (i).

The triple $(\Omega, \mathcal{F}, P)$ is the complete probabilistic model. For finite Ω with $\mathcal{F} = 2^{\Omega}$, specifying P is equivalent to specifying the numbers $P(\{\omega\})$ for each outcome ω , non-negative and summing to 1; then $P(E) = \sum_{\omega \in E} P(\{\omega\})$.

2.3 Example — fair coin

$\Omega = \{H, T\}$ and $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$ — all four subsets. The measure $P(\emptyset) = 0$, $P(\{H\}) = P(\{T\}) = \frac{1}{2}$, $P(\{H, T\}) = 1$ satisfies the axioms; note additivity is what forces $P(\{H, T\}) = P(\{H\}) + P(\{T\}) = 1$, so the only free choice was the single number $P(\{H\})$.

2.4 Example — images as sample spaces, and the counting

A 5×5 binary image is a point of $\Omega = \{0, 1\}^{5 \times 5}$, so $|\Omega| = 2^{25} \approx 3.4 \times 10^7$. The event collection is the full power set, and a set with N elements has exactly 2^N subsets (each

element is independently in or out), so $|\mathcal{F}| = 2^{2^{25}}$ — already unimaginably large for a toy image.

Notation warning (from the slides, worth internalizing): $[0, 1]$ is the real interval, an uncountable set; $\{0, 1\}$ is a two-element set. The probability measure maps into $[0, 1]$; binary pixels live in $\{0, 1\}$.

The same pattern gives the table on the slides: MNIST has $\Omega = \{0, \dots, 255\}^{28 \times 28}$ with $|\Omega| = 256^{784} = 2^{6272} \approx 10^{1888}$; CIFAR-10 has $\Omega = \{0, \dots, 255\}^{3 \times 32 \times 32}$ with $|\Omega| = 2^{24576} \approx 10^{7398}$; ImageNet at 224×224 RGB has $|\Omega| = 2^{1204224} \approx 10^{362508}$. (Conversion: $2^k = 10^{k \log_{10} 2}$ with $\log_{10} 2 \approx 0.30103$.)

Omitted remark — why the sizes matter. The number of atoms in the observable universe is roughly 10^{80} . So no dataset can ever "cover" even a vanishing fraction of Ω , and a uniform distribution over Ω would make every dataset useless: a uniformly random pixel array is noise with overwhelming probability. Real data distributions concentrate enormous mass on a tiny, structured subset (the "natural images"). That concentration is exactly the structure a model tries to learn, and it is why generative modeling is possible at all despite the astronomical ambient space.

3 Random Variables

3.1 Definition

A **random variable** is a map $X : \Omega \rightarrow \mathbb{R}$: it converts an outcome into a number. (The rigorous definition adds the measurability requirement that $\{\omega : X(\omega) \leq t\} \in \mathcal{F}$ for every $t \in \mathbb{R}$, so that probabilities of statements about X are defined. With $\mathcal{F} = 2^\Omega$ on a discrete space, every map is measurable, so the condition is automatic in this lecture.)

Notation. Uppercase X is the random variable (a function); lowercase x is a realization (a number). " $X = x$ " abbreviates the event $\{\omega \in \Omega : X(\omega) = x\}$.

The deck's two examples: on $\Omega = \{0, 1\}^{5 \times 5}$, let $X(\omega)$ be the number of 1-pixels in ω (the displayed digit "2" has $X(\omega) = 11$); on $\Omega = \{H, T\}^7$, let X count occurrences of the pattern HT , so $X(HTHTHHH) = 2$. Both illustrate that a random variable deliberately discards information: many distinct outcomes share the same value.

3.2 pmf

The **probability mass function** (pmf) of X is

$$p_X(x) = \Pr(X = x) = P(\{\omega : X(\omega) = x\}).$$

We write $\mathcal{X}$ for the set of values X can take (the **alphabet** or **support**). Since the events $\{X = x\}$, $x \in \mathcal{X}$, partition Ω , additivity gives $\sum_{x \in \mathcal{X}} p_X(x) = 1$, and $p_X(x) \geq 0$ always. Conversely, any such collection of numbers is the pmf of some random variable. In this sense the pmf **completely characterizes** the random variable: any question about probabilities of X -values is answered by p_X alone, with the underlying $(\Omega, \mathcal{F}, P)$ no longer needed. From Section 4 on, we work exclusively with pmfs.

3.3 Expectation and the law of the unconscious statistician

The **expectation** of a discrete random variable X is the probability-weighted average

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x \cdot p_X(x),$$

e.g. $\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$ for a fair die. (For infinite alphabets the sum must converge absolutely for the expectation to be well-defined; all sums in this lecture do.)

The deck leaves the following as an exercise with a hint. Here is the full solution, since the identity is used constantly — entropy itself is defined through it.

Proposition (LOTUS — law of the unconscious statistician). For any function $f : \mathcal{X} \rightarrow \mathbb{R}$,

$$\mathbb{E}[f(X)] = \sum_{x \in \mathcal{X}} f(x) p_X(x).$$

Proof. Following the hint, define the new random variable $Y = f(X)$, with alphabet $\mathcal{Y} = f(\mathcal{X})$. Its pmf is obtained by additivity over the disjoint events $\{X = x\}$, $x \in f^{-1}(y)$:

$$p_Y(y) = \Pr(f(X) = y) = \sum_{x: f(x)=y} p_X(x).$$

Applying the definition of expectation to Y and substituting:

$$\mathbb{E}[f(X)] = \mathbb{E}[Y] = \sum_{y \in \mathcal{Y}} y p_Y(y) = \sum_{y \in \mathcal{Y}} \sum_{x: f(x)=y} y p_X(x) = \sum_{y \in \mathcal{Y}} \sum_{x: f(x)=y} f(x) p_X(x) = \sum_{x \in \mathcal{X}} f(x) p_X(x),$$

where the third equality uses $y = f(x)$ inside the inner sum, and the last holds because the sets $f^{-1}(y)$, $y \in \mathcal{Y}$, partition $\mathcal{X}$ — every x appears in the double sum exactly once. ■

The name is a joke: people apply the formula "unconsciously", as if it were the definition of $\mathbb{E}[f(X)]$, when the definition actually requires first finding the pmf of $f(X)$. The proposition says the shortcut is legitimate.

4 Entropy

4.1 Surprisal

Setting for the rest of the lecture: X is a discrete random variable with pmf p_X on a finite alphabet, canonically $\mathcal{X} = \{0, 1, \dots, M-1\}$ (except where an infinite alphabet is explicitly allowed, as in the geometric exercise below).

Definition (surprisal). The surprisal of X at the outcome x is

$$S(x) = \log \frac{1}{p_X(x)} = -\log p_X(x).$$

The motivating example: let X indicate a serious earthquake ($X = 1$) versus a normal day ($X = 0$). Observing $X = 0$, an outcome with $p_X(0) \approx 1$, carries essentially no surprise; observing the rare $X = 1$ is very surprising. Surprise should be large exactly when $p_X(x)$ is small, which $-\log p_X(x)$ delivers: it is 0 at $p = 1$ and tends to $+\infty$ as $p \rightarrow 0^+$.

4.2 Why the logarithm?

Three properties single out $\log(1/p)$ among decreasing functions of p :

- **Certainty is unsurprising:** $p = 1 \Rightarrow S = 0$.
- **Monotonicity:** S is strictly decreasing in p .
- **Independent surprises add:** if two independent events have probabilities p and q , the pair has probability pq , and

$$\log \frac{1}{pq} = \log \frac{1}{p} + \log \frac{1}{q}.$$

The logarithm is essentially the only continuous function turning products into sums (Cauchy's functional equation), so additivity of information across independent observations forces the log.

Omitted remark (from the source notes). There is a deeper axiomatic justification:

Shannon's original paper [2] and later Khinchin-style axiom systems show that continuity + monotonicity in M for uniform distributions + a grouping/recursivity axiom characterize $H(X) = -\sum p \log p$ up to the base of the logarithm. This course does not develop the axiomatics. Other information measures exist and are useful, notably the **Rényi entropy** of

order $\alpha \geq 0$, $\alpha \neq 1$:

$$H_\alpha(X) = \frac{1}{1-\alpha} \log \sum_{x \in \mathcal{X}} p_X(x)^\alpha,$$

which recovers Shannon entropy in the limit $\alpha \rightarrow 1$ [4].

4.3 Entropy: definition and conventions

Definition (entropy). The entropy of X is its expected surprisal:

$$H(X) = \mathbb{E}[S(X)] = \mathbb{E}[-\log p_X(X)] = \sum_{x \in \mathcal{X}} p_X(x) \log \frac{1}{p_X(x)}.$$

Note the middle expression applies LOTUS with $f(x) = -\log p_X(x)$ — the pmf appears both as the weighting and inside the function being averaged. Two conventions:

- **Base 2; unit "bits".** All logs are base 2 unless stated. Base e gives "nats"; since $\log_2 t = \ln t / \ln 2$, entropies convert by the constant factor $1/\ln 2 \approx 1.4427$, and every theorem in this lecture is base-independent.
- **$0 \cdot \log 0 = 0$.** Terms with $p_X(x) = 0$ are defined to contribute 0, equivalently the sum runs only over the support $\{x : p_X(x) > 0\}$. This is not arbitrary; it is continuity:

$$\lim_{t \rightarrow 0^+} t \log \frac{1}{t} = 0, \quad \text{since} \quad \lim_{t \rightarrow 0^+} t \ln t = \lim_{t \rightarrow 0^+} \frac{\ln t}{1/t} \stackrel{\text{L'Hôpital}}{=} \lim_{t \rightarrow 0^+} \frac{1/t}{-1/t^2} = \lim_{t \rightarrow 0^+} (-t) = 0.$$

So an impossible outcome adds nothing to the uncertainty, and H is a continuous function of the pmf on the probability simplex, boundary included. This convention quietly matters in several proofs below (equality conditions, the endpoints of h_2).

Two readings of the same number. Entropy is the uncertainty about X before observing it, and equally the amount of information gained by observing it — resolving uncertainty and acquiring information are the same event viewed from opposite sides of the observation.

4.4 Worked examples, fully computed

Fair coin. $p_X(H) = p_X(T) = \frac{1}{2}$: both surprisals equal $\log 2 = 1$, so

$$H(X) = \frac{1}{2} \log 2 + \frac{1}{2} \log 2 = 1 \text{ bit.}$$

One fair coin toss is the unit of uncertainty.

Biased coin. $p_X(0) = 0.01$, $p_X(1) = 0.99$:

$$H(X) = 0.01 \log \frac{1}{0.01} + 0.99 \log \frac{1}{0.99} = 0.01 \times 6.644 + 0.99 \times 0.0145 \\ \approx 0.0664 + 0.0144 \approx 0.081 \text{ bits.}$$

The rare outcome is individually very surprising ($\log 100 \approx 6.64$ bits) but carries weight only 0.01; the common outcome is nearly unsurprising. Both terms are small — for different reasons — so a heavily biased coin is nearly deterministic in entropy terms. Comparing the two coins answers the slide's question "which is more random?": $H = 1$ bit versus $H \approx 0.081$ bits agrees with the obvious intuition that the fair coin is far more uncertain.

Binary entropy function. For $X \sim \text{Bernoulli}(p)$,

$$h_2(p) = -p \log p - (1 - p) \log(1 - p).$$

Basic facts, all visible in the slide's plot: $h_2(0) = h_2(1) = 0$ (using the $0 \log 0$ convention); h_2 is symmetric, $h_2(p) = h_2(1 - p)$, since swapping the labels of the two outcomes cannot change uncertainty; and h_2 peaks at $p = \frac{1}{2}$ with $h_2(\frac{1}{2}) = 1$ (a special case of the maximum-entropy theorem below, or of the concavity computed in Section 5.2). E.g. $h_2(0.01) \approx 0.081$ as above.

Fair die. Uniform on 6 faces, $p_X(x) = \frac{1}{6}$: every surprisal is $\log 6$, so

$$H(X) = \sum_{x=1}^6 \frac{1}{6} \log 6 = \log 6 \approx 2.585 \text{ bits.}$$

In general a uniform distribution on M values has entropy exactly $\log M$ — the constant surprisal makes the average trivial. This is the case of equality in Theorem 3.

Three outcomes. $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ has surprisals $(1, 2, 2)$ bits, so

$$H(X) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 2 = 1.5 \text{ bits,}$$

between the fair coin (1 bit) and the uniform distribution on 3 symbols ($\log 3 \approx 1.585$ bits). Dyadic pmfs like this one (all masses powers of $\frac{1}{2}$) are exactly those whose surprisals are integers; they will reappear when entropy is connected to optimal code lengths [1, Ch. 5].

4.5 The guessing game

Alice picks a number uniformly from $\{0, \dots, 7\}$; Bob asks yes/no questions. Compare:

- **Q1:** "Is your number in $\{0, 1, 2, 3\}$?" — answer is yes with probability $\frac{1}{2}$;
- **Q2:** "Is your number in $\{3, 6\}$?" — answer is yes with probability $\frac{1}{4}$.

The answer to each question is itself a binary random variable, and its entropy measures how much Bob expects to learn:

$$H(\text{answer to Q1}) = h_2\left(\frac{1}{2}\right) = 1 \text{ bit},$$

$$H(\text{answer to Q2}) = h_2\left(\frac{1}{4}\right) = \frac{1}{4} \cdot 2 + \frac{3}{4} \log \frac{4}{3} \approx 0.811 \text{ bits}.$$

Q1 is the more informative question. Intuitively, a question whose answer Bob can already half-guess (Q2 is "no" three times out of four) teaches him less on average. A perfectly balanced question extracts the maximum 1 bit per answer.

Twenty-questions view. Halving questions shrink the candidate set $8 \rightarrow 4 \rightarrow 2 \rightarrow 1$, so $\log 8 = 3$ balanced questions always identify Alice's number — and $H(X) = \log 8 = 3$ bits for uniform Alice. This is the operational meaning of entropy previewed here and made precise by source-coding theory: entropy is the average number of optimal yes/no questions (equivalently, code bits) needed to determine X [1, Ch. 5]. For non-uniform X the statement holds up to less than one extra question per symbol, with the dyadic case exact.

The tree the deck draws. Balanced questions build a binary tree: the root holds all eight candidates, every question splits each node in half, and after three levels each leaf holds a single number. Bob's answers trace one root-to-leaf path — for Alice's 5: "in $\{1, 2, 3, 4\}$?" no, "in $\{5, 6\}$?" yes, "is it 5?" yes — so the three answers form a 3-bit string naming the leaf. Each question is worth exactly one bit in two senses that agree here: the answer is $\text{Bernoulli}(\frac{1}{2})$, whose entropy is $h_2(\frac{1}{2}) = 1$; and the candidate set halves, $8 \rightarrow 4 \rightarrow 2 \rightarrow 1$.

Converse (why no strategy can do better). A deterministic strategy asking k yes/no questions produces at most 2^k distinct answer strings. If it always identifies X , distinct outcomes must produce distinct strings, so $2^k \geq |\mathcal{X}| = 8$, i.e. $k \geq \log 8 = 3$.

An unbalanced question does strictly worse on average: its answer has entropy $h_2(q) < 1$ for $q \neq \frac{1}{2}$, so it removes less than one bit of the $\log 8$ bits of uncertainty and more questions are needed in expectation.

4.6 Two dice

Two pmfs on the same six faces:

- $D_1 = \left(\frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{2}{9}, \frac{2}{9}, \frac{2}{9}\right)$ — mild non-uniformity;
- $D_2 = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}\right)$ — strongly concentrated (and dyadic).

$$H(D_1) = 3 \cdot \frac{1}{9} \log 9 + 3 \cdot \frac{2}{9} \log \frac{9}{2} = \frac{1}{3} \log 9 + \frac{2}{3} \log 4.5$$

$$\approx \frac{1}{3}(3.170) + \frac{2}{3}(2.170) \approx 1.057 + 1.447 \approx 2.503 \text{ bits},$$

$$H(D_2) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{16} \cdot 4 + \frac{1}{32} \cdot 5 + \frac{1}{32} \cdot 5 = \frac{8+8+6+4+\frac{5}{4}+\frac{5}{4}}{16} = \frac{31}{16} \approx 1.938 \text{ bits}.$$

D_1 is the more uncertain die: it is closer to uniform. Both entropies sit below $\log 6 \approx 2.585$, foreshadowing the maximum-entropy theorem.

The deck plots both pmfs beneath the entropy axis, each tied to its tick. D_2 is a halving staircase — one face carries half the mass — and lands at 1.94; D_1 is nearly flat (three bars at $\frac{1}{9}$, three at $\frac{2}{9}$) and lands at 2.50, just under the uniform ceiling $\log 6$. Visual flatness is exactly what the number scores.

4.7 AI examples

Classifier soft labels. A digit classifier over labels $\{0, \dots, 4\}$ outputs a pmf $f(x)$ — a soft guess, like a weather forecast. For two images,

$$f(x_1) = (0.6, 0.1, 0.1, 0.1, 0.1) : \quad H(f(x_1)) = 0.6 \log \frac{1}{0.6} + 4 \times 0.1 \log 10$$

$$\approx 0.442 + 1.329 \approx 1.771 \text{ bits},$$

$$f(x_2) = (0.4, 0.4, 0.2, 0, 0) : \quad H(f(x_2)) = 2 \times 0.4 \log 2.5 + 0.2 \log 5$$

$$\approx 1.058 + 0.464 \approx 1.522 \text{ bits}$$

(the two zero entries of $f(x_2)$ contribute nothing, by the $0 \log 0$ convention). So entropy ranks x_1 as the harder sample: its mass is spread over all five labels, while $f(x_2)$ has effectively a three-label support. Note the comparison is not obvious by eye — $f(x_2)$ has the smaller top probability gap between its two leaders — which is precisely why a scalar summary is useful.

Data valuation / active learning. "Which unlabeled sample is most helpful to train on next?" One classical answer: score each candidate by the entropy of the model's current soft label and prioritize high-entropy (most-confused) samples — these are the ones the model can learn most from (uncertainty sampling; see [6]).

Exploration vs exploitation — where the tension comes from. Strip reinforcement learning down to its smallest interesting case, the K -armed bandit [8]. There are K strategies ("arms"); arm a pays a random reward with unknown mean μ_a . At each round t the learner picks A_t and observes one noisy sample R_t with $\mathbb{E}[R_t \mid A_t = a] = \mu_a$. Performance is measured by regret against the best fixed arm,

$$\mathcal{R}_T = T \max_a \mu_a - \sum_{t=1}^T \mathbb{E}[\mu_{A_t}].$$

The tension is structural: the only way to learn μ_a is to play a . After n_a pulls the estimate $\hat{\mu}_a$ carries a confidence width of order $\sqrt{\log T/n_a}$, which shrinks only for arms the learner keeps choosing. A purely greedy learner — always play $\arg \max_a \hat{\mu}_a$ — can therefore lock onto a suboptimal arm permanently: one unlucky early sample of the genuinely best arm B drags $\hat{\mu}_B$ down, B is never pulled again, and the error is never corrected. That is the deck's picture. Arm A has 40 pulls and a tight interval around a good value; arm B has 3 pulls and an interval so wide that its upper end sits above A 's estimate. B might be better, and no amount of thinking settles it — only rounds spent on B do.

Both sides of the trade cost something. Pulling B gives up the reward A would reliably have paid (the exploitation loss, bounded and one-off). Never pulling B risks a fixed per-round shortfall $\mu_B - \mu_A$ for the rest of time (linear regret). Lai and Robbins showed the trade-off cannot be dodged: any strategy that is uniformly good must pull each suboptimal arm $\Omega(\log T)$ times [9], and UCB-style rules — play the arm with the largest $\hat{\mu}_a + \sqrt{2 \log t/n_a}$, i.e. be optimistic in the face of uncertainty — match that rate [10]. ϵ -greedy and Thompson sampling are the other standard answers. Full RL adds states and delayed rewards on top, but the dilemma is already complete here.

This is where entropy enters. Rather than hand-tuning when to deviate, modern deep-RL objectives add an entropy bonus $\beta H(\pi(\cdot | s))$ to the reward (entropy regularization in A3C/PPO; soft actor-critic maximizes reward plus policy entropy [11]). A high-entropy policy keeps probability on every action, so every action's estimate keeps improving — exploration becomes a term in the objective instead of a heuristic.

Exploration in RL. With strategies $\{A, B, C, D, E\}$ and current belief "A is best", compare two stochastic policies $S_1 = (0.6, 0.1, 0.1, 0.1, 0.1)$ and $S_2 = (0.4, 0.4, 0.2, 0, 0)$. These are numerically the same pmfs as above, so $H(S_1) \approx 1.771 > H(S_2) \approx 1.522$: S_1 is the more explorative policy — it reserves probability for every option, while S_2 can never try D or E . Entropy is used exactly this way as an exploration bonus/regularizer in modern RL.

4.8 Exercise — geometric distribution, solved

Let X be geometric with $p = \frac{1}{2}$: $p_X(x) = 2^{-x}$ for $x \in \{1, 2, 3, \dots\}$ (this is a valid pmf: $\sum_{x \geq 1} 2^{-x} = 1$). This is the first infinite-alphabet example; the entropy sum is an infinite series and must be checked for convergence.

The surprisal of outcome x is $\log 2^x = x$ bits — exactly the outcome itself. Hence

$$H(X) = \sum_{x=1}^{\infty} p_X(x) \cdot x = \mathbb{E}[X].$$

It remains to evaluate $\mathbb{E}[X] = \sum_{x \geq 1} x 2^{-x}$. Let $T = \sum_{x=1}^{\infty} x 2^{-x}$ (the series converges, e.g. by the ratio test). Then

$$T - \frac{1}{2}T = \sum_{x=1}^{\infty} x 2^{-x} - \sum_{x=1}^{\infty} x 2^{-(x+1)} = \sum_{x=1}^{\infty} x 2^{-x} - \sum_{x=2}^{\infty} (x-1) 2^{-x} = \sum_{x=1}^{\infty} 2^{-x} = 1,$$

so $\frac{1}{2}T = 1$, i.e. $\mathbb{E}[X] = 2$ and $\boxed{H(X) = 2 \text{ bits}}$. ■

Remark. An infinite alphabet with finite entropy — and note $\log |\mathcal{X}| = \infty$ here, so the maximum-entropy bound of Section 5 is vacuous (it genuinely requires finite M to say anything). There even exist pmfs on $\{1, 2, \dots\}$ with $H(X) = \infty$ (e.g. $p_X(x) \propto 1/(x \log^2 x)$), so finiteness of entropy on infinite alphabets is a real hypothesis, not a formality.

5 Properties of Entropy

5.1 Theorem 1 — non-negativity

Theorem 1. $H(X) \geq 0$, with equality if and only if X is deterministic (i.e. $p_X(x^*) = 1$ for some single value x^*).

Proof. Every probability satisfies $0 \leq p_X(x) \leq 1$, so $\frac{1}{p_X(x)} \geq 1$ on the support and every surprisal is $\geq \log 1 = 0$. An expectation of a non-negative random variable is non-negative:

$$H(X) = \mathbb{E}\left[\log \frac{1}{p_X(X)}\right] \geq \mathbb{E}[\log 1] = 0.$$

Equality. $H(X) = \sum_x p_X(x) \log \frac{1}{p_X(x)}$ is a sum of non-negative terms, so it vanishes iff every term vanishes: $p_X(x) \log \frac{1}{p_X(x)} = 0$ for all x , i.e. every $p_X(x) \in \{0, 1\}$ (the factor $p_X(x)$ kills the term at 0 — the $0 \log 0$ convention again — and the log kills it at 1; strictly between, both factors are strictly positive). Since the masses sum to 1 and each is 0 or 1, exactly one value x^* has $p_X(x^*) = 1$: X is deterministic. Conversely a deterministic X has the single term $1 \cdot \log 1 = 0$. ■

So zero entropy characterizes "no randomness at all". Next question: which pmf sits at the top of the scale? The plot of h_2 suggests: uniform. Proving it needs a tool.

5.2 Background — convexity and concavity

Definition. A function $f : I \rightarrow \mathbb{R}$ on an interval I is **convex** if for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(\lambda x + (1 - \lambda)y),$$

and **concave** if the reverse inequality holds ($\leq$). Geometrically: convex means every chord lies on or above the graph; concave, on or below. f is strictly convex/concave if the inequality is strict whenever $x \neq y$ and $0 < \lambda < 1$. Note f is concave iff $-f$ is convex.

The slides check membership by picture (x^2 convex, $\log x$ concave, h_2 concave). The deck's hint is the second-derivative test, which we now state and prove, since both Jensen's equality analysis and the exercises rely on it.

Lemma (second-derivative test). If f is twice differentiable on an interval and $f''(t) \leq 0$ throughout, then f is concave; if $f''(t) < 0$ throughout, f is strictly concave. (Signs reversed for convexity.)

Proof. Fix $x < y$, $\lambda \in (0, 1)$, and let $z = \lambda x + (1 - \lambda)y$, so $z - x = (1 - \lambda)(y - x)$ and $y - z = \lambda(y - x)$, both positive. By the mean value theorem there are $\xi_1 \in (x, z)$ and $\xi_2 \in (z, y)$ with

$$f(z) - f(x) = f'(\xi_1)(z - x), \quad f(y) - f(z) = f'(\xi_2)(y - z).$$

Since $f'' \leq 0$, f' is non-increasing and $\xi_1 < \xi_2$ gives $f'(\xi_1) \geq f'(\xi_2)$. Therefore

$$\lambda(f(z) - f(x)) = \lambda(1 - \lambda)(y - x)f'(\xi_1) \geq \lambda(1 - \lambda)(y - x)f'(\xi_2) = (1 - \lambda)(f(y) - f(z)),$$

and rearranging yields $f(z) \geq \lambda f(x) + (1 - \lambda)f(y)$: concavity. If $f'' < 0$ then f' is strictly decreasing, the middle inequality is strict, and f is strictly concave. ■

The key example. $f(t) = \log t$ on $(0, \infty)$ has $f''(t) = -\frac{1}{t^2 \ln 2} < 0$: the logarithm is strictly concave. This single fact powers Theorem 3, and (next lecture) KL non-negativity.

Exercise from the deck, solved — h_2 is concave. For $p \in (0, 1)$, differentiate $h_2(p) = -p \log p - (1 - p) \log(1 - p)$:

$$h_2'(p) = -\log p - \frac{1}{\ln 2} + \log(1 - p) + \frac{1}{\ln 2} = \log \frac{1 - p}{p},$$

$$h_2''(p) = \frac{1}{\ln 2} \left(-\frac{1}{1 - p} - \frac{1}{p} \right) = -\frac{1}{p(1 - p) \ln 2} < 0.$$

So h_2 is strictly concave on $(0, 1)$; by continuity (with the endpoint convention) it is concave on $[0, 1]$. As a byproduct, $h_2'(p) = 0$ iff $p = \frac{1}{2}$, confirming the peak at the fair coin. The convexity of x^2 is the same test: $(x^2)'' = 2 > 0$.

5.3 Theorem 2 — Jensen's inequality

Theorem 2 (Jensen [3]). Let f be concave and X a random variable taking finitely many values in the domain of f . Then

$$\mathbb{E}[f(X)] \leq f(\mathbb{E}[X]).$$

For convex f the inequality reverses. If f is strictly concave, equality holds if and only if X is constant (all its probability mass on one point).

How to read it. Concavity compares a two-point mixture with the function value at the two-point average; Jensen upgrades this to an arbitrary pmf: the weighted average of function values, $\mathbb{E}[f(X)] = \sum_i p_i f(x_i)$, never exceeds the function evaluated at the weighted average input, $f(\sum_i p_i x_i)$. Jensen is thus the "many-point" generalization of the definition — and the proof is exactly an induction that reduces many points to two.

Proof (induction on the support size M). Write the claim for a pmf $(p_1, \dots, p_M)$ on points $(x_1, \dots, x_M)$:

$$\sum_{i=1}^M p_i f(x_i) \leq f\left(\sum_{i=1}^M p_i x_i\right).$$

Base case $M = 1$: both sides equal $f(x_1)$. Base case $M = 2$: with $p_2 = 1 - p_1$, the claim reads

$$p_1 f(x_1) + (1 - p_1) f(x_2) \leq f(p_1 x_1 + (1 - p_1) x_2),$$

which is precisely the definition of concavity with $\lambda = p_1$.

Induction step. Assume the claim for every pmf on $M - 1$ points; take a pmf on M points. If $p_M = 1$ the claim is trivial (it is the $M = 1$ case), so assume $p_M < 1$ and peel off the M -th atom:

$$\sum_{i=1}^M p_i f(x_i) = p_M f(x_M) + (1 - p_M) \sum_{i=1}^{M-1} q_i f(x_i), \quad q_i = \frac{p_i}{1 - p_M}.$$

The renormalized weights satisfy $q_i \geq 0$ and $\sum_{i=1}^{M-1} q_i = \frac{1 - p_M}{1 - p_M} = 1$: a valid pmf on $M - 1$ points. Now apply the induction hypothesis to the inner sum, then two-point concavity (with $\lambda = p_M$) to recombine:

$$\begin{aligned} p_M f(x_M) + (1 - p_M) \sum_{i=1}^{M-1} q_i f(x_i) &\leq p_M f(x_M) + (1 - p_M) f\left(\sum_{i=1}^{M-1} q_i x_i\right) \\ &\leq f\left(p_M x_M + (1 - p_M) \sum_{i=1}^{M-1} q_i x_i\right) = f\left(\sum_{i=1}^M p_i x_i\right), \end{aligned}$$

the last equality because $(1 - p_M) q_i = p_i$. This is $\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$. ■

Equality for strictly concave f (the part the slides state without proof). If X is constant, both sides are f of that constant — equality. Conversely, suppose X takes two distinct values $x \neq y$ with positive probabilities. Group the outcomes: run the argument with atom masses $p = \Pr(X = x) > 0$ on x and $1 - p$ on the conditional mixture of everything else. Tracing the induction, the final recombination step applies two-point concavity to two distinct points with interior weight, which is strict for strictly concave f ; hence $\mathbb{E}[f(X)] < f(\mathbb{E}[X])$. (Alternative slick proof for the differentiable case: strict concavity gives the supporting-line bound $f(t) \leq f(m) + f'(m)(t - m)$ with equality only at $t = m$, where $m = \mathbb{E}[X]$; take expectations of both sides. This one-line version is worth remembering.)

Why we care. Today: maximum entropy. Next lecture: $D(P\|Q) \geq 0$. Throughout ML: whenever an expectation is trapped under a log (log-likelihoods, ELBO), Jensen moves the expectation inside at the cost of an inequality in a known direction.

5.4 Theorem 3 — maximum entropy

Theorem 3. For a random variable X on a finite alphabet $\mathcal{X}$ with $|\mathcal{X}| = M$,

$$H(X) \leq \log |\mathcal{X}| = \log M,$$

with equality if and only if X is uniform on $\mathcal{X}$, i.e. $p_X(x) = \frac{1}{M}$ for all x .

Proof. A subtlety first: we may assume $p_X(x) > 0$ for all $x \in \mathcal{X}$. If some values had zero mass, restrict to the support $\mathcal{X}' = \{x : p_X(x) > 0\}$: by the $0 \log 0$ convention H is unchanged, and $\log |\mathcal{X}'| \leq \log M$, so proving the bound on the support proves it on $\mathcal{X}$. (This also shows equality forces full support — see below.)

Apply Jensen to the random variable $\frac{1}{p_X(X)}$, which is positive, using the (strictly) concave function $\log$:

$$H(X) = \mathbb{E} \left[\log \frac{1}{p_X(X)} \right] \leq \log \mathbb{E} \left[\frac{1}{p_X(X)} \right].$$

Now evaluate the inner expectation by LOTUS (Section 3.3), with $f(x) = \frac{1}{p_X(x)}$ — the pmf cancels itself:

$$\mathbb{E} \left[\frac{1}{p_X(X)} \right] = \sum_{x \in \mathcal{X}} p_X(x) \cdot \frac{1}{p_X(x)} = \sum_{x \in \mathcal{X}} 1 = M.$$

Combining, $H(X) \leq \log M$. ■

Equality. The only inequality used was Jensen with the strictly concave $\log$, which is tight iff the random quantity inside is constant: $\frac{1}{p_X(x)} = c$ for all x in the support, i.e. $p_X(x) = \frac{1}{c}$ there. Summing to 1 over the support forces $c = |\text{support}|$; and to reach the full bound $\log M$ (rather than $\log$ of a smaller support) the support must be all of $\mathcal{X}$, giving $p_X(x) = \frac{1}{M}$ for every x : the uniform distribution. Conversely, the uniform distribution attains $H = \log M$ by the fair-die computation. ■

Sanity checks against every example of the lecture. Fair coin: $H = 1 = \log 2$, tight (uniform). Biased coin: $0.081 < 1$. Dice: $H(D_1) \approx 2.503$ and $H(D_2) \approx 1.938$, both below $\log 6 \approx 2.585$, and D_1 (closer to uniform) is closer to the ceiling. Fair die: $\log 6$ exactly, tight. Soft labels: 1.771 and 1.522, both below $\log 5 \approx 2.322$. The theorem also explains the guessing game: a uniform Alice is the hardest opponent — $\log M$ is simultaneously the number of bits needed to index M items and the entropy of the uniform pick, and any non-uniform strategy of Alice's would lower the entropy and give Bob structure to exploit.

Scale of uncertainty. Theorems 1 and 3 bracket every pmf on M symbols:

$$0 \leq H(X) \leq \log M,$$

with the deterministic distributions at the floor and the uniform distribution alone at the ceiling. Entropy turns "how random is this pmf?" into a position on one axis.

5.5 Bridge to Lecture 2

The surprisal $\log \frac{1}{p_X(x)}$ uses the true pmf. A learned model only has a guess Q , and one can ask how $\mathbb{E}[\log \frac{1}{Q(X)}]$ (the cross-entropy of Q against p_X) compares to $H(X)$. The source notes prove, by the same Jensen pattern applied to $\mathbb{E}[\log \frac{Q(X)}{p_X(X)}]$, that the mismatched description is never cheaper: $\mathbb{E}[\log \frac{1}{Q(X)}] \geq H(X)$ with equality iff $Q = p_X$, and the gap defines the KL divergence $D(p_X \| Q) \geq 0$. That theorem, its $p_X(x) = 0$ edge case, and the gambling interpretation are the content of Lecture 2 (prob02-kl-crossentropy) and its companion note.

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